% Experiment 1: Generation of Continuous-Time Signals clc; clear; close all;

% Time Vector t = -1:0.001:1;

% Parameters A = 2; f0 = 5; a = 1;

% Signal Generation imp = double(abs(t) < 0.0005); step = double(t >= 0); ramp = t .* step; expSig = exp(-a*t) . step; sinSig = A * sin(2*pi*f0*t);

% Plotting Signals figure;

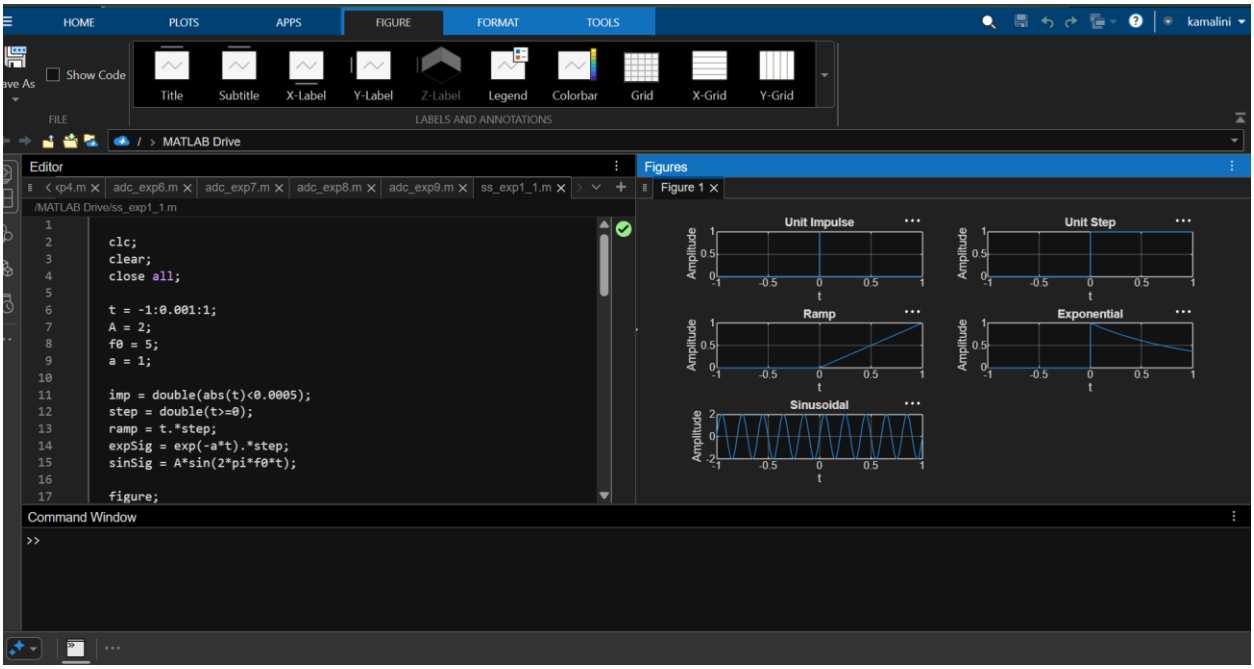
subplot(3,2,1); plot(t, imp); grid on; title('Unit Impulse'); xlabel('Time (s)'); ylabel('Amplitude');

subplot(3,2,2); plot(t, step); grid on; title('Unit Step'); xlabel('Time (s)'); ylabel('Amplitude');

subplot(3,2,3); plot(t, ramp); grid on; title('Ramp Signal'); xlabel('Time (s)'); ylabel('Amplitude');

subplot(3,2,4); plot(t, expSig); grid on; title('Exponential Signal'); xlabel('Time (s)'); ylabel('Amplitude');

subplot(3,2,5); plot(t, sinSig); grid on; title('Sinusoidal Signal'); xlabel('Time (s)'); ylabel('Amplitude');



Objective 2 :

% Experiment 2: Generation of Discrete-Time Signals clc; clear; close all;

% Parameters n = -20:20; A = 2; f0 = 0.1; a = 0.1;

% Signal Generation imp = double(n == 0); step = double(n >= 0); ramp = n .* step; expSig = (0.8).^n .* step; sinSig = A * sin(2*pi*f0*n);

% Plotting Signals figure;

subplot(3,2,1); stem(n, imp, 'filled'); grid on; title('Unit Impulse'); xlabel('n'); ylabel('Amplitude');

subplot(3,2,2); stem(n, step, 'filled'); grid on; title('Unit Step'); xlabel('n'); ylabel('Amplitude');

subplot(3,2,3); stem(n, ramp, 'filled'); grid on; title('Ramp Signal'); xlabel('n'); ylabel('Amplitude');

subplot(3,2,4); stem(n, expSig, 'filled'); grid on; title('Exponential Signal'); xlabel('n'); ylabel('Amplitude');

subplot(3,2,5); stem(n, sinSig, 'filled'); grid on; title('Discrete-Time Sinusoid'); xlabel('n'); ylabel('Amplitude');

